Theorem-Backed Static Cost Forecasting with Refutable Explanations for Quantum Compilation

Sze Chun Yiu
Independent Researcher
sze-chun.yiu@fysik.su.se

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Abstract

A compilation-cost predictor can be numerically correct for three very different reasons: it may be exact by theorem, empirically accurate on a benchmark, or correct only because a compact regime classifier happened to cover the observed inputs. We separate these certificate layers for a frozen shared-Tag TARE grammar. The initial closed-form forecaster $F_1(t) = \min(C_{R6L}, C_{D^+}, f_B)$ is inexpensive and interpretable, and prospectively staged tests confirm 102/102 predicted rows, but a later frozen $n = 3$ instance exactly refutes it: $C_{DP} = 10$ while $F_1 = 11$. The repair is theorem-guided rather than post-hoc curve fitting. An existing all-n support theorem proves $C_{DP} = C_{D^{++}}$ under the unit support-count objective, so we define $F_2(t) = C_{D^{++}}(t)$ using an independent support-≤ 2 evaluator with no unrestricted-DP call. F_2 has zero error on 9,547 compared instances, including the refuting row; its exactness comes from the theorem, while the benchmark checks the implementation. A compact enlarged-borrow explanation B' also repairs the observed row, but a later hostile QG-7 search finds 64 exact hybrid configurations outside B'. Crucially, those explanation failures remain inside D^{++} and therefore do not affect F_2 cost exactness. A further hybrid family closes 10,481 verified explanation rows, while one all-n classification lemma remains open. The resulting design principle is certificate separation: exact cost, regime explanation, prospective validation, and verification status should be reported independently. The result is scoped to the frozen grammar/objective and does not imply full-circuit or hardware resource optimality.

Keywords: quantum resource estimation; certified forecasting; exact compilation; proof-carrying computation; regime classification.

1 Introduction

Resource estimation often sits between a fast analytical formula and an expensive optimizer. A formula is attractive because it can be evaluated before compilation, but its authority is easy to overstate. Zero error on a benchmark is not a theorem. Conversely, a theorem about a broad exact family does not automatically provide a compact explanation of which structural regime produced a cost.

The TARE regime programme offers a useful separation. An initial three-family expression had a clear semantic interpretation: donor-exact, split-anchor, or central borrow. It matched exhaustive and held-out panels and was used for prospective staging. QG-5 then did what a useful certification system should allow: it found one exact fresh counterexample and recorded the forecast as wrong rather than weakening the gate.

The later support-two theorem makes a stronger repair possible. Instead of adding a case to the old formula and calling the fit complete, we evaluate the exact support-≤ 2 family whose equality with unrestricted DP is already proved for all n under the frozen objective. This produces a theorem-backed exact cost forecaster F_2. Compact regime explanations can then be developed and refuted independently on top of that cost layer.

This paper argues for that layered view of a certified forecast. We distinguish the cost theorem, the implementation check against DP, the human-readable regime model, prospective prediction evidence, and row-level verification authority. The QG-7 refutation of the first repaired explanation is valuable precisely because the layered contract prevents it from being mistaken for a failure of the exact cost certificate.

2 Methods

2.1 Forecast target and truth referee

The target is the exact frozen R6M structural support-count cost $C_{DP}(t)$. Unrestricted dynamic programming is used as a finite referee where executed, but it is not called in the forecast path. All statements are indexed to the unit objective and frozen grammar.

2.2 Certificate layers

We report four layers separately. **Cost certificate**: a theorem or exact containment argument that the predicted cost equals C_{DP}. **Explanation certificate**: a structural family/predicate assigning a donor/trade regime. **Empirical validation**: comparison on frozen referee rows. **Verification authority**: whether a particular library row has a committed exact-referee receipt. Failure of a weaker layer does not automatically invalidate a stronger one.

2.3 Initial closed form

The QG-5 forecaster uses the frozen envelope

$$F_1(t) = \min\{C_{R6L}(t), C_{D^+}(t), f_B(t)\}.$$

Each term is computable without unrestricted DP and corresponds to a named structural configuration. Its completeness was machine-evidenced on earlier panels, not proved all-n.

2.4 Theorem-backed support-two forecaster

R6S proves $C_{DP} = C_{D^{++}}$ for every n under the unit objective. QG-5b therefore defines

$$F_2(t) = C_{D^{++}}(t),$$

computed by a frozen independent support-≤ 2 enumerator. The implementation is checked against committed DP truth and proof-carrying witnesses. For high-n chemistry rows, equality may be certified by the same exact containment pinch used in the underlying receipts rather than a direct D++ sweep.

2.5 Prospective and hostile evaluation

QG-3 freezes predicted rows before referee access. QG-5 freezes a new random panel and permits forecast refutation. QG-7 later freezes adversarial witness shapes against the enlarged-borrow explanation B' without changing its definition after outcome inspection. This sequence tests calibration under both supportive and hostile data.

3 Results

3.1 Prospective successes do not establish global exactness

QG-3 stages 102 predictions before exact evaluation and confirms all 102: 90 real-library matching rows are donor-exact, and 12 engineered boundary rows cover four predicted donor, four split, and four borrow configurations. This is strong prospective evidence that the original classifier can make correct positive regime predictions. It is still finite evidence.

3.2 QG-5 produces the required refutation

The frozen QG-5 benchmark compares F_1 with exact truth on 9,546 rows. It is exact on the 9,261 structured-$n = 2$ rows and the committed chemistry rows, but one fresh seeded $n = 3$ instance has $C_{DP} = 10 < 11 = F_1$. The witness uses a support-two frame whose effective borrow home lies outside that block's own target support. The error is serialized rather than removed from the panel.

This refutation localizes the weakness: the support-two theorem is not challenged, but the closed-form borrow family is too narrow. The result therefore distinguishes 'forecast formula incomplete' from 'support theorem false'.

3.3 QG-5b restores theorem-backed exact cost

QG-5b evaluates $F_2 = C_{D^{++}}$. All integrity gates pass, including no unrestricted-DP call in the forecast path and independent witness referees. F_2 has zero cost error on 9,547 compared instances. On the QG-5 counterexample it returns 10, matching both the unrestricted referee and a reverified D^{++} witness.

The benchmark agreement is not used as the logical proof of exactness. R6S already proves $C_{DP} = C_{D^{++}}$ for all n in the frozen grammar/objective. The 9,547 rows verify that the implemented evaluator and receipt bindings behave consistently with that theorem.

3.4 A repaired explanation is refuted without breaking the cost certificate

QG-5b also enlarges the borrow family to B' by permitting out-of-support/union-support phantom homes. On the verified QG-5b panels, the repaired closed-form envelope matches exact cost and correctly reclassifies the old false-positive row.

QG-7 then freezes a hostile search against exactly that family and finds 64 fourth-configuration witnesses among 740 evaluated instances. Each has $C_{D^{++}} < \min(C_{D^+}, f_{B'})$, with gap -1, while $C_{DP} = C_{D^{++}}$ still holds. A representative mechanism combines a weight-two Tag with a phantom borrow. Thus B' is refuted as an all-n explanation but F_2 remains exact.

QG-7b adds a frozen hybrid family B'' and closes 10,481 verified rows with no fifth configuration on those panels. QG-7c closes the high-Tag-weight rung and most remaining shapes, leaving one pinned comm-$s2$ lemma sector open. The explanation certificate is therefore improving but not yet an all-n theorem.

3.5 Verification authority is row-specific

Library forecast rows with committed DP receipts can be labeled exact/verified under their bound. Rows emitted for eligible candidates without referee receipts retain verification authority 'NONE'. The table is therefore simultaneously a prediction product and an honesty check: printing a number does not turn it into verified evidence.

4 Discussion

The refutation sequence supports a practical rule for scientific resource estimators: attach authority to components, not to the word 'forecast'. F_2 has an all-n cost certificate under the frozen objective. A donor/split/borrow/hybrid label has weaker, evolving authority. A library row may have no exact-referee verification even when the underlying cost theorem applies structurally. These distinctions should remain visible in any downstream resource table.

The QG-5 failure also shows why prospective successes are not a substitute for hostile evaluation. The original formula had exhaustive $n = 2$ evidence and 102/102 prospective staged confirmations, yet a single $n = 3$ row defeated its completeness. The correct response was to preserve the counterexample and move to the theorem-backed family, not to average the error away.

QG-7 provides an even stronger calibration test. It refutes the human-readable repaired explanation while leaving the exact cost layer untouched. In other words, explanation can lag prediction. For some applications that is acceptable: an exact cost with a weaker mechanism label may still be useful, provided the certificate explicitly says which part is theorem-backed.

The remaining engineering question is efficiency. A closed form may be much cheaper than a support-two enumerator, while both are cheaper/different from unrestricted DP in different regimes. Runtime should therefore be measured for the final canonical implementation and reported separately from correctness; this manuscript does not use speed as part of the exactness claim.

5 Related work and boundary

The forecasted object is built on an established Pauli/block-encoding substrate. The parent TARE construction is a stabilizer-formalism method for block encoding linear combinations of Pauli strings [1]. Pauli-intermediate-representation compiler optimization is well represented by frameworks such as PCOAST [2], and recent Hamiltonian-simulation compilers perform global binary-symplectic simplification over Pauli structure [3]. These works support the neighboring compilation context; they are not cited as evidence that the particular cost certificate or certificate hierarchy developed here is already known.

The paper sits more broadly between quantum resource estimation, analytical cost modeling, certified optimization, proof-carrying computation, and static prediction. Its bounded candidate residual is the explicit certificate layering demonstrated by the TARE refutation/repair chain: a theorem-backed exact restricted-family cost evaluator, a separately falsifiable structural explanation, prospectively staged predictions, hostile exact-referee tests, and row-level verification authority. A fresh literature pass must test whether an existing resource-estimation or certified-compiler framework already packages these components equivalently.

All underlying TARE, Pauli/symplectic, dynamic-programming, and donor-family machinery remains donor-owned or is credited to the companion technical results. The claim of this paper is about forecasting and certification structure, and its empirical demonstration.

Conflated cost, and what separating it buys. The concern that a single compilation-cost number hides distinct resources is not new. Sarkar argues that quantum compilers optimize execution-only proxies — gate count, depth, fidelity — and that treating the compiled circuit as the unit of cost conflates two different things: how much the substrate must do at run time, and how much must be said to describe what to do [4]. Two circuits with identical gate count, depth and fidelity are not interchangeable artifacts if one is a loop and the other its unrolled expansion. That work separates cost into an execution axis and a description axis, and shows the joint optimum can differ from either.

The separation pursued here is orthogonal to that one, and the distinction matters. Sarkar separates *what is being counted*; this paper separates *why the count is believed.* A forecast can be numerically right because a theorem forces it, because it fit a benchmark, or because a regime classifier happened to cover the observed inputs — three different warrants behind one correct number, distinguishable only by what would refute each. The two decompositions are compatible: a description-aware cost model would still admit the certificate question about each of its axes.

Precision is not the only axis. Resource estimation already recognises that a cost number can be untrustworthy, and organises the problem by precision. Campbell et al. describe the field as split between fast, hardware-agnostic formula estimates and slow, application-specific analyses tuned to one architecture, and observe that the former neglect implementation detail while the latter cannot be reconfigured when assumptions change; their compilation-driven framework aims between the two [5]. Their diagnosis of the failure mode is explicit: existing approaches rely on fixed architectural assumptions or coarse analytical models that miss the interaction between hardware constraints and compilation.

The axis separated here is different, and the two are independent. A more precise estimator is still an estimator whose correctness rests on something, and that something is not recorded by the estimate. The forecaster studied here can be exactly right because a theorem forces the value, because it was fitted to a benchmark that happened to cover the inputs, or because a compact regime classifier covered them — three warrants that a precision-oriented account does not distinguish, because all three can produce the same number to the same accuracy. What separates them is not error magnitude but what would refute each: a theorem-backed value is refuted by a single exact counterexample, a benchmark fit by held-out inputs, a regime classifier by an input outside its regime. A compilation-driven estimator of the kind Campbell et al. build would still admit that question about each of its own layers.

6 Limitations

F2's all-n exactness is objective- and grammar-specific. QG-2 supplies a reweighted objective where support three can pay, so the R6S certificate cannot be transferred. Other TARE ranks, other block-encoding families, or full hardware cost vectors require new proofs.

'Static' does not mean constant-time or closed-form. F2 evaluates a bounded support-two family with an independent enumerator. The paper makes no universal runtime-superiority claim over unrestricted DP; implementation/runtime comparisons should be remeasured on the final code/environment.

The compact regime explanation remains incomplete all-n. QG-7b's zero gap on 10,481 rows and QG-7c's near-closed proof chain are finite/partial evidence, not a theorem. Exact cost without exact explanation may be insufficient for applications that require mechanistic attribution.

Chemistry coverage is limited to named receipted batches, and some higher-n equalities use exact containment pinches rather than direct support-two sweeps. Prediction tables for unrefereed candidates are not validation data.

7 Conclusion

The TARE forecasting sequence shows how a certified estimator should behave under refutation. An interpretable three-family formula passes extensive and prospective tests, then fails on one exact fresh instance. A theorem-backed support-two evaluator repairs the cost layer and is exact for all n under the frozen unit objective. Later hostile search refutes the repaired compact explanation again without refuting the exact cost.

The scientific contribution is therefore not a perfect closed form. It is a layered certificate: theorem-backed cost, separately scoped explanation, prospective/hostile empirical checks, and row-specific verification authority. This structure makes counterexamples informative rather than catastrophic and prevents benchmark accuracy from being mistaken for proof.

8 Reproducibility

The primary chain is committed under `research/extensions/orion-qg/` with frozen protocols under `development/orion-qg-regime-geometry/`. QG-3 supplies prospective staging; QG-5 supplies the refuted F_1 benchmark; QG-5b supplies F2 and the enlarged-borrow repair; QG-7/QG-7b/QG-7c supply the hostile explanation stress tests. R6S under the parent ORION-Q tree is the all-n cost theorem.

A reproduction must preserve the chronology. It should first re-establish the old forecast failure, then verify the theorem-backed F2 result, then independently reproduce the later B' explanation refutation. Collapsing those stages into a final zero-error table would erase the scientific evidence that the certificate separation is meant to expose.

The root `REPRODUCE.md` lists the required checks and the distinction between direct referee rows, theorem-backed rows, and unverified prediction rows.

9 Ethics, safety and resource statement

No human participants, animals, clinical data, or personal data are used. The relevant integrity risk is epistemic over-certification: reporting a benchmark-accurate formula as proved, reporting an exact cost theorem as an exact regime explanation, or presenting an unrefereed library prediction as verified. The layered certificate and explicit verification-authority fields are designed to prevent these promotions.

Computation ranges from closed-form/restricted-family evaluation to exact dynamic programming and hostile finite searches. Resource use is reported separately from correctness. The protected stretched-N2 subject remains outside the evidence, and no forecast here grants physical quantum-advantage or deployment authority.

References

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